4. Collaboration and Remote Repositories:
Clone a remote Git repository to your local machine.

1. Aim
To clone a remote Git repository to a local machine using Git commands and understand how to work with remote repositories.

2. Learning objectives
After completing this lab program, will be able to:

- Understand the concept of remote repositories.
- Clone a repository from GitHub to a local computer.
- Navigate the cloned repository using the terminal.
- Verify the repository status and remote URL.
- Understand the relationship between local and remote repositories.

3. Prerequisites
Before starting the program, ensure that:

1. Git is installed on your computer.
2. You have an internet connection.
3. You have a GitHub account if you want to clone your own private repository. A public repository can be cloned without signing in.
4. You have a repository URL to clone.

To verify Git installation, open Command Prompt and type:

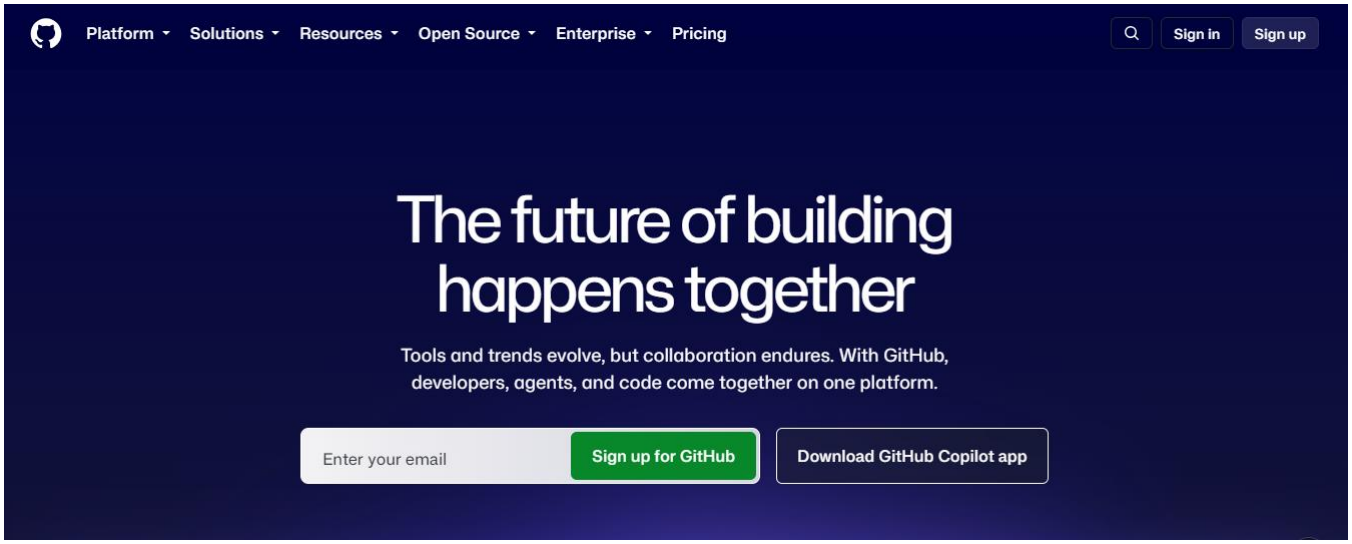
```
git --version
```
Example output:

```
git version 2.55.0.windows.5
```
The version number may differ depending on your installation.

If Git is not installed, download it from the official website: https://git-scm.com/downloads

4. Step-by-step procedure

Step 1: Open GitHub



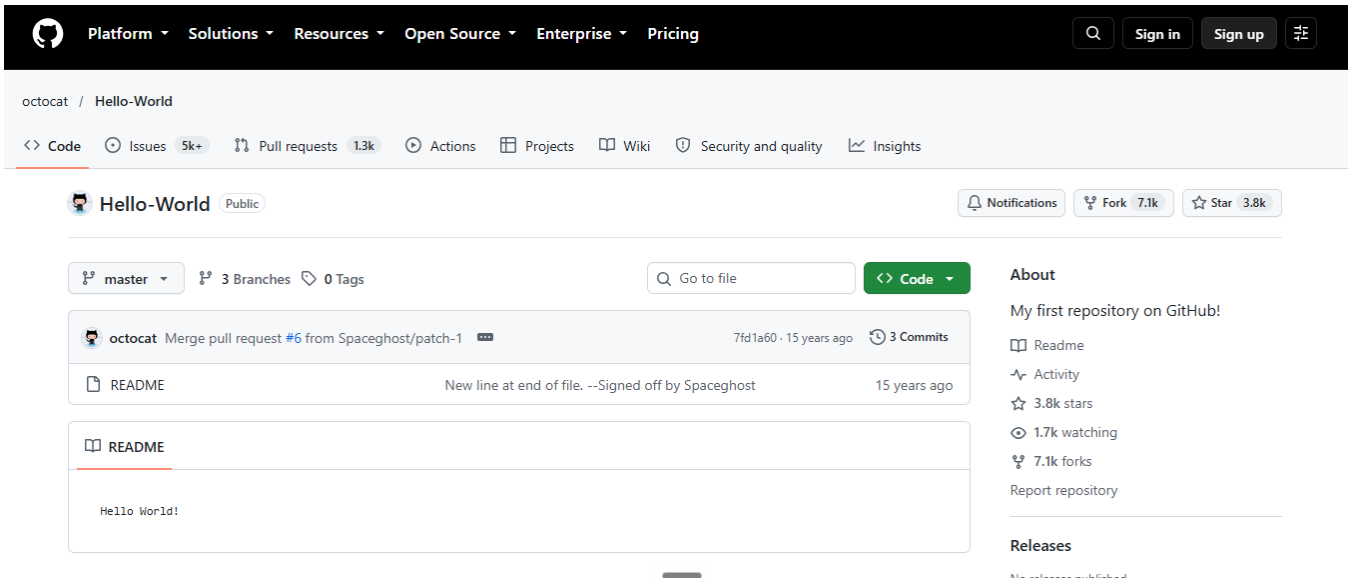
1. Open a web browser.
2. Visit https://github.com.
3. Sign in to your GitHub account if required.
4. Open the repository you want to clone.

For demonstration, assume that the repository URL is:

```
https://github.com/octocat/Hello-World.git
```

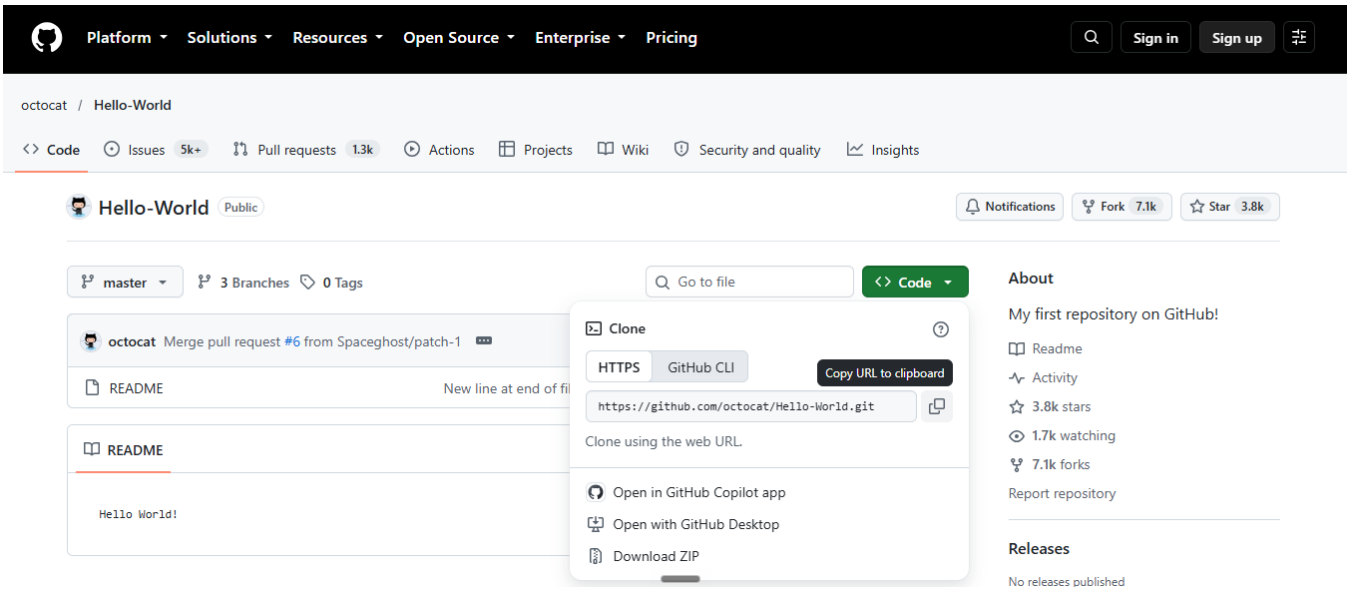
This is a sample public repository. You may use your own repository URL for the lab.

Step 2: Copy the repository URL



5. Open the required repository.
6. Click the Code button.
7. Select the HTTPS option.

8. Click the copy icon next to the repository URL.



The copied URL will look similar to:

https://github.com/octocat/Hello-World.git

Explanation: The repository URL identifies the remote Git repository hosted on GitHub.

Step 3: Open the terminal
Open **Command Prompt**

Suppose you want to store the cloned repository inside a folder called **GitPg4**. First, navigate to the desired location. For example, in Windows Command Prompt:

Create a folder for your lab work:

```
mkdir GitPg4
```
Enter the folder:

```
cd GitPg4
```
Explanation:

- cd changes the current directory.
- mkdir creates a new directory.

Step 4: Clone the remote repository
Execute the following command:

```
git clone https://github.com/octocat/Hello-World.git
```

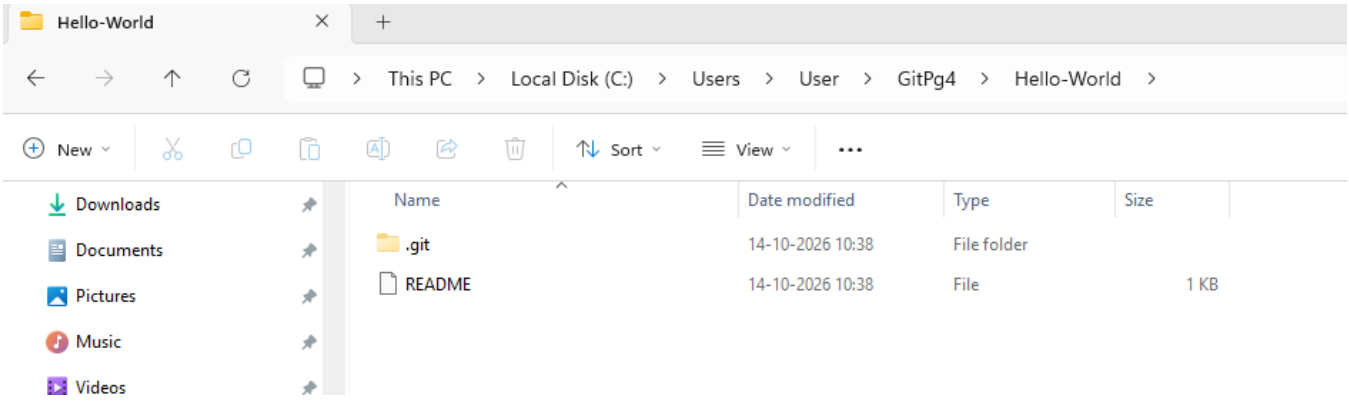
Command explanation
git clone downloads a repository from a remote location and creates a local working copy, including its Git history and configuration for the remote.

```
C:\Users\User\GitPg4>git clone https://github.com/octocat/Hello-World.git
Cloning into 'Hello-World'...
remote: Enumerating objects: 13, done.
remote: Total 13 (delta 0), reused 0 (delta 0), pack-reused 13 (from 1)
Receiving objects: 100% (13/13), done.
```

What happens internally?

1. *Git connects to the remote repository.*
2. *It downloads the repository's available Git history and data.*
3. *It creates a new local directory named Hello-World.*
4. *It checks out the repository's default branch.*
5. *It configures the remote repository under the name origin.*

The downloaded files will be stored inside the Hello-World directory.



Step 5: Enter the cloned repository

Run:

`cd Hello-World`

Explanation: This command moves into the local repository directory created by git clone.

To view the files and folders, use:

`dir`

```
C:\Users\User\GitPg4\Hello-World>dir
 Volume in drive C has no label.
 Volume Serial Number is A2D1-9D67

 Directory of C:\Users\User\GitPg4\Hello-World

14-10-2026  10:38    <DIR>          .
14-10-2026  10:38    <DIR>          ..
14-10-2026  10:38                14 README
               1 File(s)             14 bytes
               2 Dir(s)   4,316,602,368 bytes free
```

You should see the files and directories that belong to the cloned repository.

Step 6: Check the repository status
Execute:

```
git status
```
Example output:

```
C:\Users\User\GitPg4\Hello-World>git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
```

The branch name may be main or another branch, depending on the repository.

Explanation:

- On branch master tells you the current branch.
- Your branch is up to date indicates that the local branch matches its tracked remote branch based on the last fetched information.
- working tree clean means there are no uncommitted changes in the working directory.

Step 7: Verify the remote repository URL
Execute:

```
git remote -v
```
Example output:

```
C:\Users\User\GitPg4\Hello-World>git remote -v
origin  https://github.com/octocat/Hello-World.git (fetch)
origin  https://github.com/octocat/Hello-World.git (push)
```

Explanation:

- **git remote -v** displays configured remote repositories and their URLs.
- origin is the default name Git assigns to the remote when cloning.
- fetch indicates the URL used to retrieve remote changes.
- push indicates the URL used to send local commits to the remote.

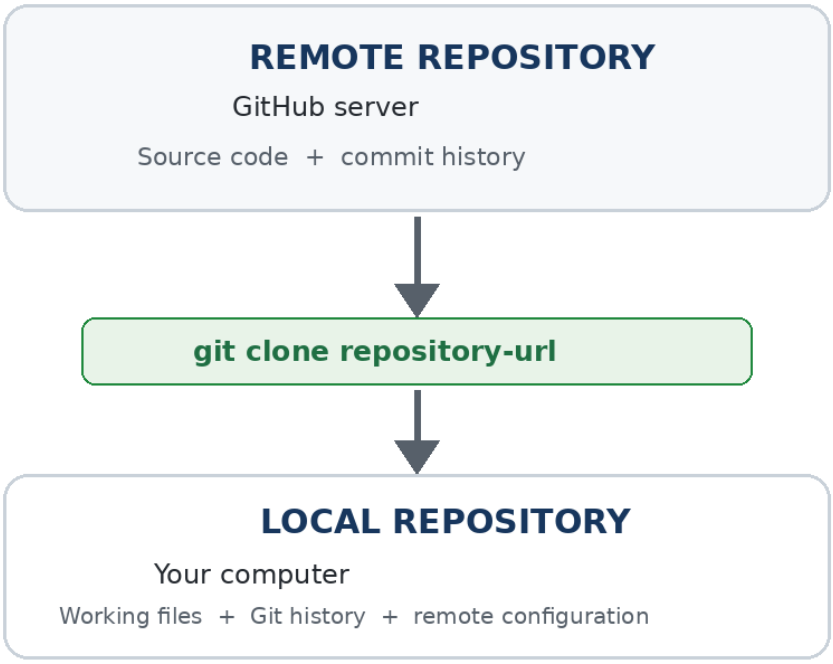
Step 8: View the commit history
Execute:

```
git log --oneline
```
Example output:

```
C:\Users\User\GitPg4\Hello-World>git log --oneline
7fd1a60 (HEAD -> master, origin/master, origin/HEAD) Merge pull request #6 from Spaceghost/patch-1
7629413 New line at end of file. --Signed off by Spaceghost
553c207 first commit
```

Explanation: This command displays the commit history in a compact format. It helps verify that the cloned repository contains its version-control history.

5. Visual representation of the process



Cloning creates a local copy of the remote repository. After cloning, you can edit files, commit changes locally, fetch updates from GitHub, and push your commits when you have the required permissions.

6. Important Git commands

Command	Purpose
git --version	Checks the installed Git version
git clone URL	Clones a remote repository
cd repository-name	Enters the cloned directory
dir or ls	Lists files and folders
git status	Displays the working tree and branch status
git remote -v	Displays remote repository URLs
git log --oneline	Displays a compact commit history

7. Expected result

The remote Git repository is successfully cloned to the local machine. The repository files and commit history are available locally, and the remote URL can be verified using git remote -v.

8. Viva questions for this program

1. What is Git?
2. What is a remote repository?
3. What is the purpose of the git clone command?
4. What is the difference between a local repository and a remote repository?

5. What is GitHub?
6. What is the purpose of the origin name?
7. What is the difference between git clone and git init?
8. What is the difference between git fetch and git pull?
9. Can you clone a public repository without a GitHub account?
10. What happens if the repository URL is incorrect?
11. What is the purpose of git status?
12. How can you verify that a repository is connected to a remote?
13. Does cloning download the commit history?
14. What is the difference between HTTPS and SSH repository URLs?
15. Can you modify files in a cloned repository without internet access?

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